

JIARUI GUO

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Email: ntguojiarui@pku.edu.cn

Homepage: <https://ntguojiarui.github.io>

GitHub: github.com/ntguojiarui

EDUCATION

Peking University

Ph.D. in Computer Science

Beijing, China

Sept. 2023 – Present

Peking University

B.Sc in Computer Science & Mathematics

Beijing, China

Sept. 2019 – Jun. 2023

WORKING EXPERIENCE

Alibaba Group

Research Intern, LLM Inference

Beijing, China

Oct. 2025 – Present

ByteDance

Research Intern, Network Infrastructure

Beijing, China

Aug. 2025 – Oct. 2025

Infinigence AI

Research Intern, System Communication

Beijing, China

Dec. 2024 – Apr. 2025

Baichuan AI

Research Intern, LLM Inference

Beijing, China

Apr. 2024 – Dec. 2024

Theory Lab, Huawei Technologies

Research Intern (Remote), LLM Training

Hong Kong SAR, China

Sept. 2023 – Apr. 2024

PUBLICATIONS

Rethinking Retrieval-Augmentation as Synthesis: A Query-Aware Context Merging Approach

Jiarui Guo, Yuemeng Xu, Zongwei Lv, Yangyujia Wang, Xiaolin Wang, Kan Liu, Tao Lan, Lin Qu, Tong Yang

arXiv preprint arXiv:2603.20286. Mar. 2026

The Molecular Structure of Thought: Mapping the Topology of Long Chain-of-Thought Reasoning

Qiguang Chen, Yantao Du, Ziniu Li, Jinhao Liu, Songyao Duan, Jiarui Guo, Minghao Liu, Jiaheng Liu, Tong Yang, Ge Zhang, Libo Qin, Wanxiang Che, Wenhao Huang

arXiv preprint arXiv:2601.06002. Jan. 2026

MagnifierSketch: Quantile Estimation Centered at One Point

Jiarui Guo, Qiushi Lyu, Yuhan Wu, Haoyu Li, Zhaoqian Yao, Yuqi Dong, Xiaolin Wang, Bin Cui, Tong Yang

arXiv preprint arXiv:2511.22070. Nov. 2025

Zephyrus: Scaling Gateways Beyond the Petabit-Era with DPU-Augmented Hierarchical Co-Offloading

Yuemeng Xu, Haoran Chen, Jiarui Guo, Mingwei Cui, Qiuhe Yin, Cheng Dong, Daxiang Kang, Xian Wu, Chenmin Sun, Peng He, Yang Gao, Lirong Lai, Kai Wang, Hongyu Wu, Tong Yang, Xiyun Xu

arXiv preprint arXiv:2510.11043. Oct. 2025

HourglassSketch: An Efficient and Scalable Framework for Graph Stream Summarization

Jiarui Guo, Boxuan Chen, Kaicheng Yang, Tong Yang, Zirui Liu, Qiuhe Yin, Sha Wang, Yuhan Wu, Xiaolin Wang, Bin Cui, Tao Li, Xi Peng, Renhai Chen, Gong Zhang

2025 IEEE International Conference on Data Engineering (ICDE '25). May. 2025

CuckooGraph: A Scalable and Space-Time Efficient Data Structure for Large-Scale Dynamic Graphs

Zhuochen Fan, Yalun Cai, Zirui Liu, Jiarui Guo, Xin Fan, Tong Yang, Bin Cui

2025 IEEE International Conference on Data Engineering (ICDE '25). May. 2025

SteadySketch: A High-Performance Algorithm for Finding Steady Flows in Data Streams

Zhuochen Fan, Xiangyuan Wang, Xiaodong Li, Jiarui Guo, Wenrui Liu, Haoyu Li, Sheng Long, Zheng Zhong, Tong Yang, Xuebin Chen, Bin Cui

IEEE/ACM Transactions on Networking (ToN). Aug. 2024

Enabling Space-Time Efficient Range Queries with REncoder

Zhuochen Fan, Bowen Ye, Ziwei Wang, Zheng Zhong, Jiarui Guo, Yuhan Wu, Haoyu Li, Tong Yang, Yaofeng Tu, Zirui Liu, Bin Cui

International Journal on Very Large Data Bases (VLDBJ). Aug. 2024

SketchPolymer: Estimate Per-item Tail Quantile Using One Sketch

Jiarui Guo, Yisen Hong, Yuhan Wu, Yunfei Liu, Tong Yang, Bin Cui

2023 ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '23). Aug. 2023

SteadySketch: Finding Steady Flows in Data Streams

Xiaodong Li, Zhuochen Fan, Haoyu Li, Zheng Zhong, Jiarui Guo, Sheng Long, Tong Yang, Bin Cui

2023 IEEE/ACM International Symposium on Quality of Service (IWQoS '23). Jun. 2023

PISketch: Finding Persistent and Infrequent Flows

Zhuochen Fan, Zhoujing Hu, Yuhan Wu, Jiarui Guo, Sha Wang, Wenrui Liu, Tong Yang, Yaofeng Tu, Steve Uhlig

IEEE/ACM Transactions on Networking (ToN). May. 2023

Finding Simplex Items in Data Streams

Zhuochen Fan, Jiarui Guo, Xiaodong Li, Tong Yang, Yikai Zhao, Yuhan Wu, Bin Cui, Yanwei Xu, Steve Uhlig, Gong Zhang

2023 IEEE International Conference on Data Engineering (ICDE '23). Apr. 2023

REncoder: A Space-Time Efficient Range Filter with Local Encoder

Ziwei Wang, Zheng Zhong, Jiarui Guo, Yuhan Wu, Haoyu Li, Tong Yang, Yaofeng Tu, Huanchen Zhang, Bin Cui

2023 IEEE International Conference on Data Engineering (ICDE '23). Apr. 2023

BurstSketch: Finding Bursts in Data Streams

Ruijie Miao, Zheng Zhong, Jiarui Guo, Zikun Li, Tong Yang, Bin Cui

IEEE Transactions on Knowledge and Data Engineering (TKDE). Nov. 2022

PISketch: Finding Persistent and Infrequent Flows

Zhuochen Fan, Zhoujing Hu, Yuhan Wu, Jiarui Guo, Wenrui Liu, Tong Yang, Hengrui Wang, Yifei Xu, Steve Uhlig, Yaofeng Tu

2022 ACM SIGCOMM Workshop on Formal Foundations and Security of Programmable Network Infrastructures (FFSPIN '22). Aug. 2022